Model Context Protocol

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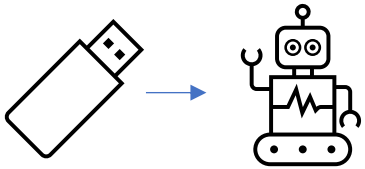
Model Context Protocol MCP)

- Open-standard
- Open-source framework
- Connects agents to tools, APIs
- Uses structured inputs/outputs
- Introduced by Anthropic in 11/2024
- Standardize how AI agents
  - Integrate data
  - Share data with external tools



USB for AI

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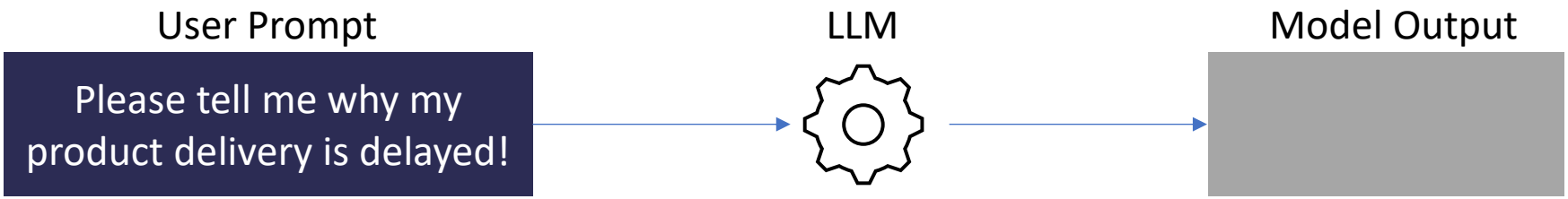
Problem



- LLM needs to know about the context to provide a good answer
- Here: context on user, its booked flight, and real-time information on flight status

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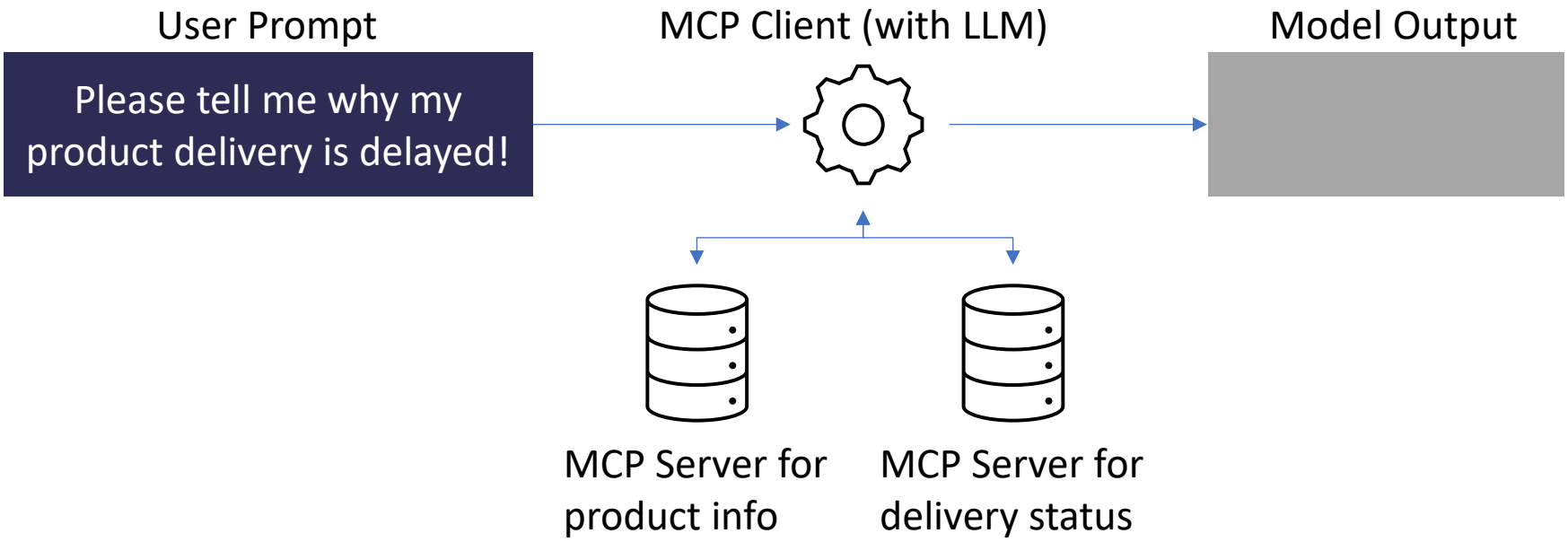
Solution



- MCP Servers provide context information

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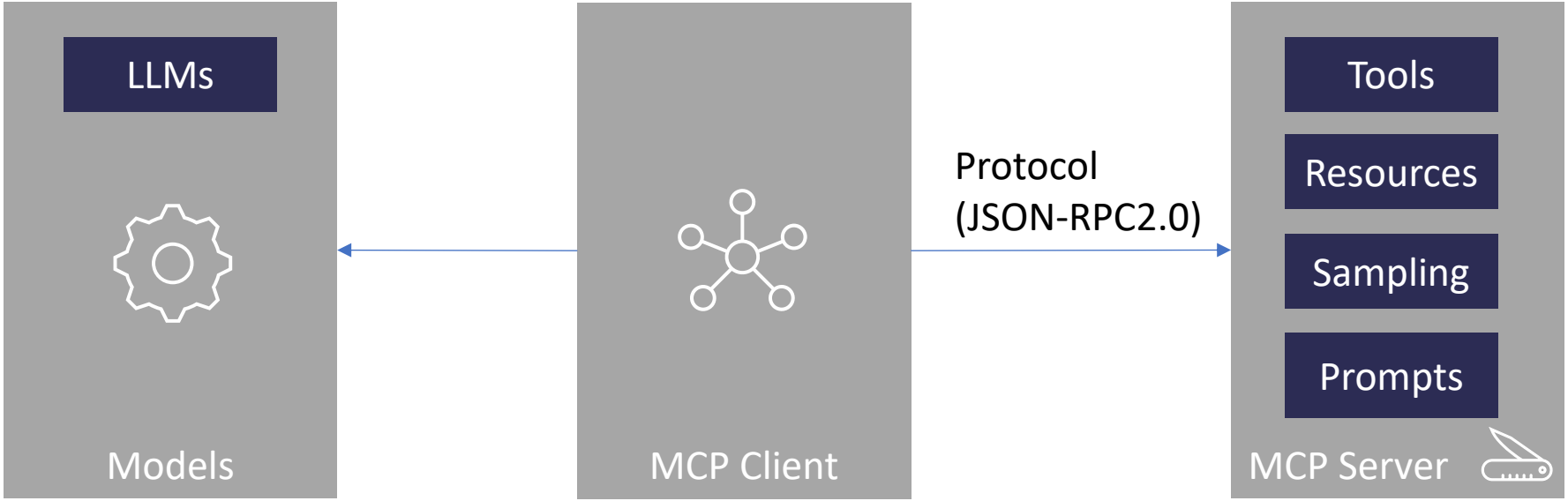
MCP Server Context Types



- foundation models
- other models

- makes the request to the server for tools and resources

- server handles requests from MCP clients

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MCP Clients

Continue
by Continue Dev, Inc.
Coding assistant plugin for VSCode and JetBrains. Highly configurable and open source.

Claude Desktop
by Anthropic
Chat with Claude. Anthropic's native implementation for macOS and Windows.

Zed
by Zed Industries
Code editor. Highly performant and open source.

Sourcegraph Cody
by Sourcegraph
Coding assistant plugin for VSCode and JetBrains.

Cline
by Cline Bot Inc.
Coding assistant plugin for VSCode with autonomous CLI capabilities. Open source.

ChatMCP
by daodao97
Chat with LLM's. UI for macOS, Windows, Linux.

HyperChat
by Big Sweet Potato Studio
Chat with LLM's for macOS and Windows. Open Source.

Sage
by Michael Latman
Chat with Claude for macOS, iOS, and iPadOS.

LibreChat
by Danny Avila
Chat with LLM's for macOS and Windows. Open source.

5ire
by UOMS
Chat with LLM's for macOS and Windows. Open source.

Systemprompt
by Ejb503
Gemini-powered MCP server control with your voice.

Toolbase
by Toolbase
Enhance Claude Desktop with baked-in, one-click toggle MCP capabilities. Open source.

CLI Inspector
by wong2
CLI tool for interacting directly with MCP servers via stdio or SSE, enabling fine-grained exploration of Tools, Resources, and Prompts.

Roo Code (prev. Roo Cline)
by Roo Vet
Fork of Cline. Coding assistant plugin for VSCode with autonomous CLI capabilities. Open source.

Goose
by Block, Inc.
Configurable software engineering AI agent that runs on your local machine.

Cursor IDE
by Anysphere
AI-powered code editor.

Source: https://www.pulsemcp.com/clients

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MCP Servers

ArXiv
John Blazick • Nov 29, 2024
Search and analyze academic papers from the arXiv repository.
65.6k 200

MCP DB Utils
Dong Hao • Feb 10, 2025
DButils is an all-in-one MCP service that enables your AI to do data analysis by accessing versatile types of database (sqlite, mysql, postgres, and more) within a unified connection configuration in a safe way.
54.6k 41

arXiv Research
Andy Brandt • Dec 7, 2024
Search and analyze scientific papers from arXiv.
52.2k 38

LLM Code Context
cyberchitta • Dec 6, 2024
Streamlines code context sharing with LLMs by implementing smart file selection, code outlining, and multi-language support for efficient code reviews and documentation generation.
49.6k 153

Crawl4AI (Web Scraping & Crawling)
Ritvij Kumar Sharma • Mar 6, 2025
Integrates web scraping and crawling capabilities with tools for single-page content extraction, metadata retrieval, and Google search functionality, enabling workflows that require web content analysis, data collection, or online research tasks.
47.8k 1

Cloudflare
Cloudflare • Nov 27, 2024
Manage KV stores, R2 storage, D1 databases, and Workers deployments.
43.6k 1k

DuckDuckGo Search
Nick Clyde • Feb 23, 2025
Integrates with DuckDuckGo to provide web search capabilities, content fetching, and parsing, with results formatted for large language model consumption.
37.3k 42

KOSPI/KOSDAQ Stock Data
Sangrok Mun • Feb 14, 2025
Integrates with pykrx to provide real-time access to KOSPI and KOSDAQ stock market data, offering tools for retrieving ticker symbols, OHLCV data, market capitalization, fundamental metrics, and trading volumes.
35.3k 16

Sentry
Anthropic • Nov 19, 2024
Retrieve and analyze application error reports for debugging support.
35.1k 26k

Source: https://www.pulsemcp.com/servers

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Protocol

- JSON-RPC
- supports reflection requests
  - client asks for tools, or resources
  - servers responds with available information

MCP Client Installation

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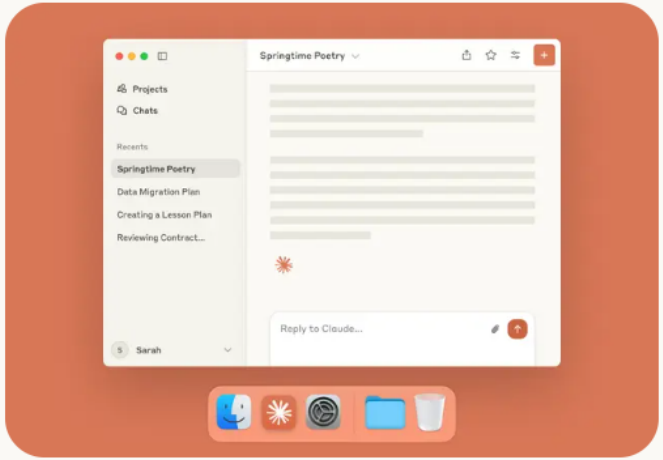
Example

Install MCP Client

Meet Claude on your desktop BETA

Always there when you need it—Claude sits quietly in your workflow. No tab-hopping required.

Windows | Windows (arm64) | macOS



Source: https://claude.ai/download

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Example

- Install MCP Python SDK: uv add "mcp[cli]"
- MCP Python SDK: https://github.com/modelcontextprotocol/python-sdk

Define Server

```python
# server.py
from mcp.server.fastmcp import FastMCP
import os

# Create an MCP server
mcp = FastMCP("Demo")


# Add an addition tool
@mcp.tool()
def list_files(folder_path: str) -> list[str]:
    """List files in a folder"""
    return os.listdir(folder_path)

@mcp.tool()
def write_markdown_file(file_path: str, content: str) -> str:
    """Write a markdown file
    Args:
        file_path: path to the file
        content: content to write to the file
    Returns:
        success message
    """
```

Review next file

server.py

Install MCP Server

```
(GenAiNewTrends) C:\Temp\GenAiNewTrends\mcp_test\markdown>mcp install  server.py
[06/25/25 15:34:15] INFO     Added server 'Demo' to Claude config                    claude.py:143
                    INFO     Successfully installed Demo in Claude app                 cli.py:504
```

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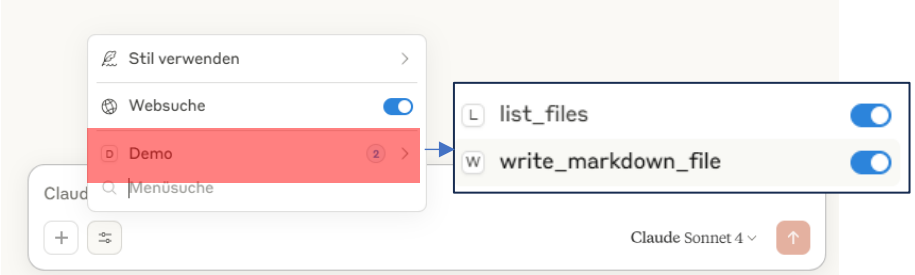
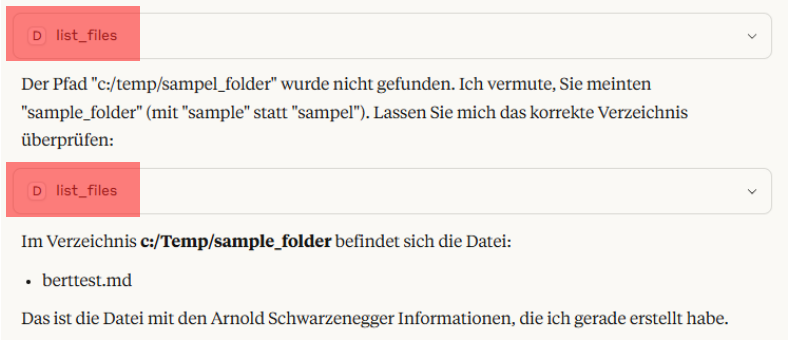
Example



MCP Functionality integrated in Claude



Claude uses the tools